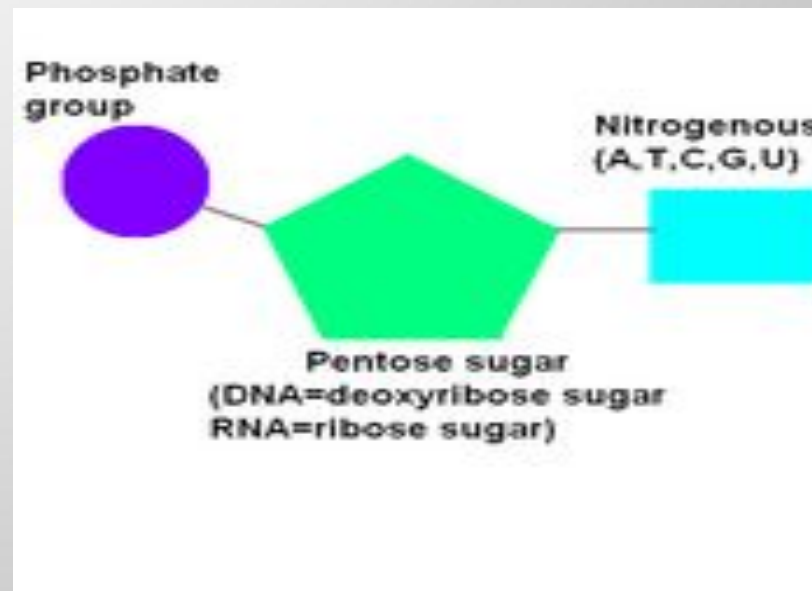


PURINE & PYRIMIDINE METABOLISM



Dr. Humaira Aman

OBJECTIVES

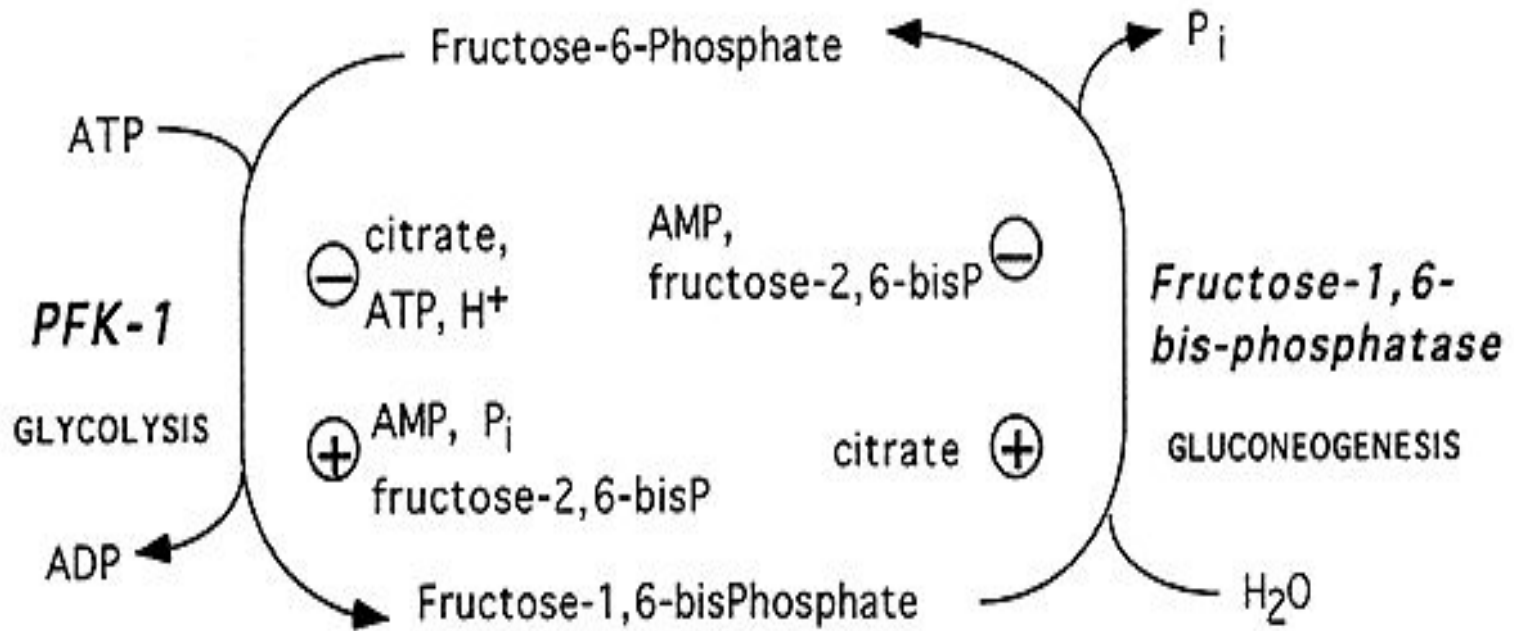
At the end of the lecture students would be able to:

- ▣ Describe briefly the structures of nitrogenous bases, nucleoside and nucleotides.
- ▣ Explain the de novo and salvage synthetic pathway of purine and pyrimidine.

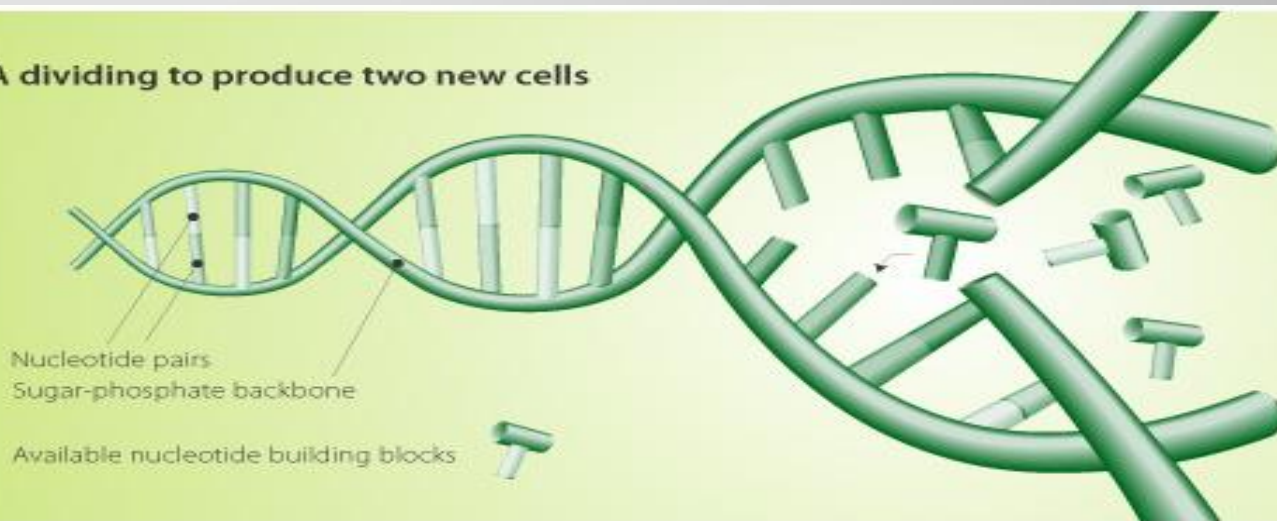
INTRODUCTION

Nucleotides are essential for all cells because:

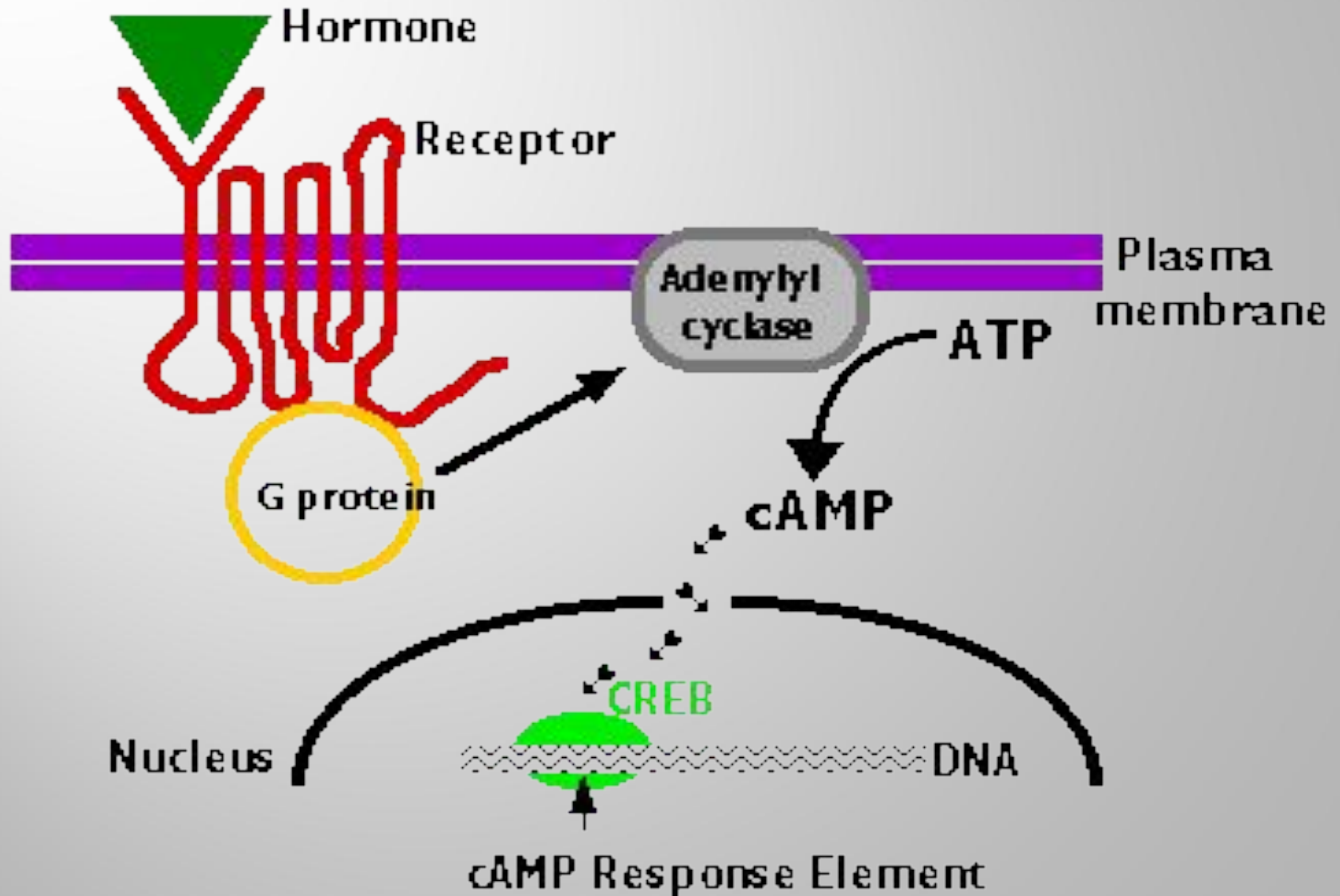
- Take part in the synthesis of nucleic acids
- Involved in the synthesis of proteins , CHO and lipids
- Structural component of coenzymes e:g coenzyme A, FAD, NAD and NADP
- Second messenger system
- Energy currency of cell
- Regulatory compounds in intermediary metabolism



DNA dividing to produce two new cells



Second Messenger System



NUCLEOTIDE STRUCTURE

Phosphate



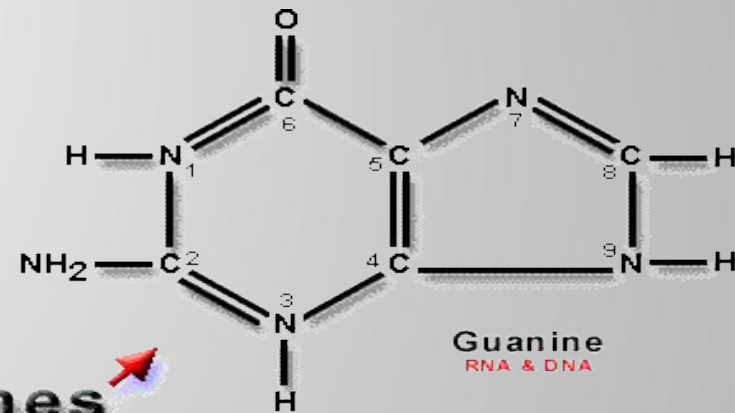
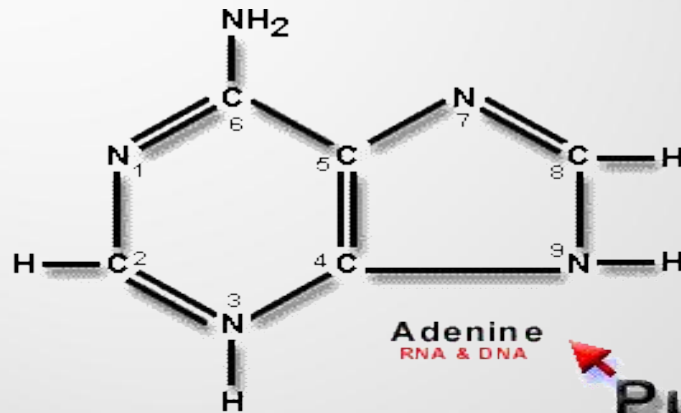
Nitrogenous Base



Pentose
Sugar

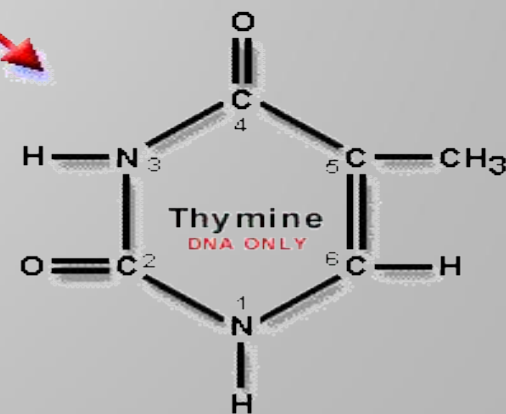
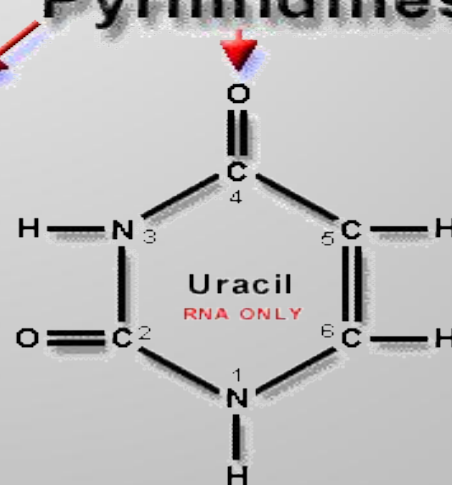
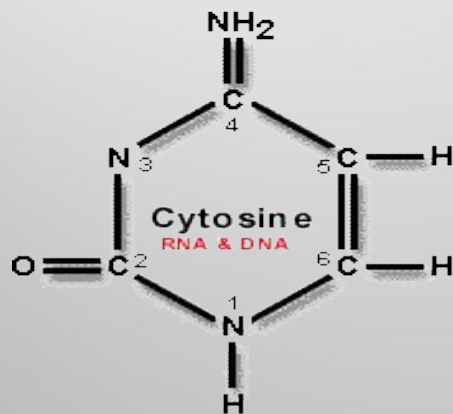


PURINE AND PYRIMIDINE STRUCTURE

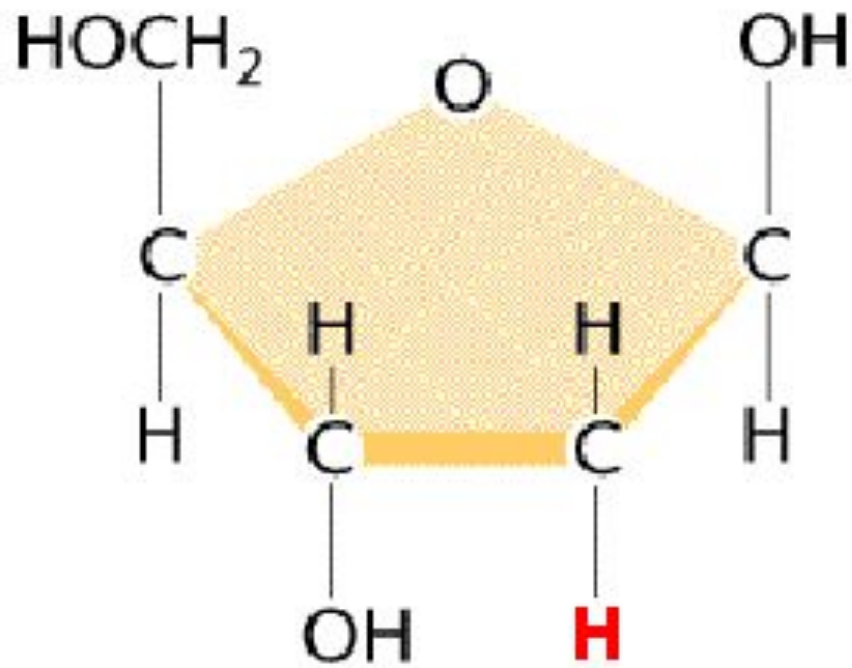


Purines

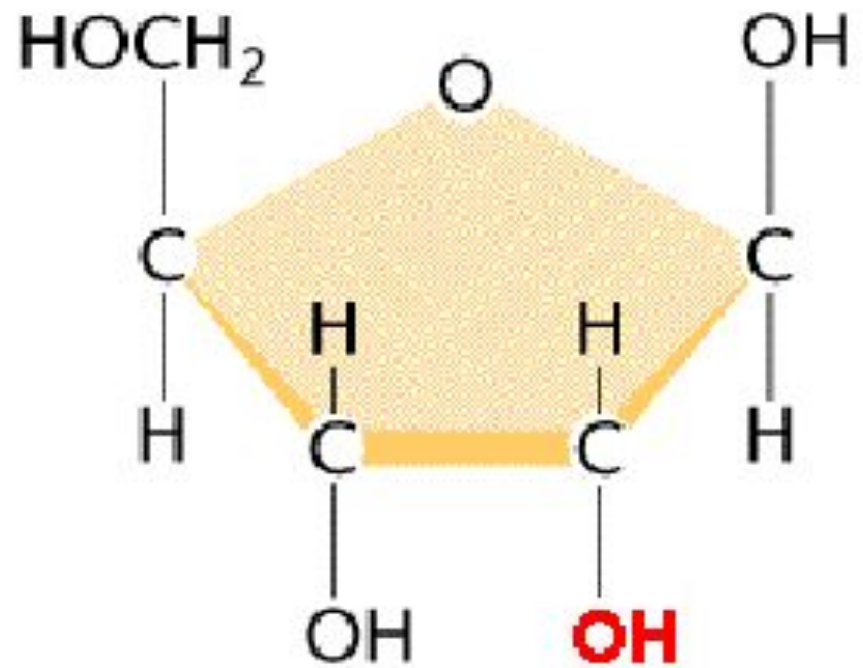
Pyrimidines



PENTOSE SUGAR STRUCTURE



2-Deoxyribose



Ribose

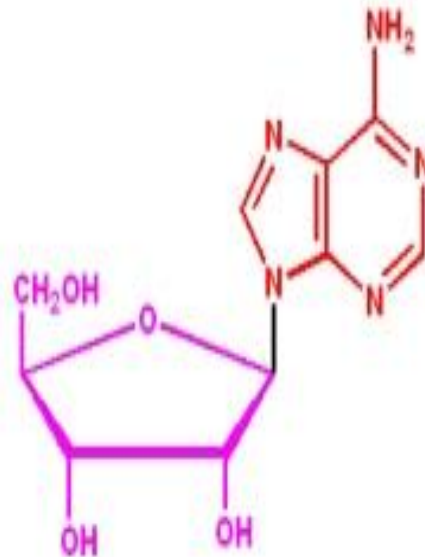
NUCLEOSIDES STRUCTURE



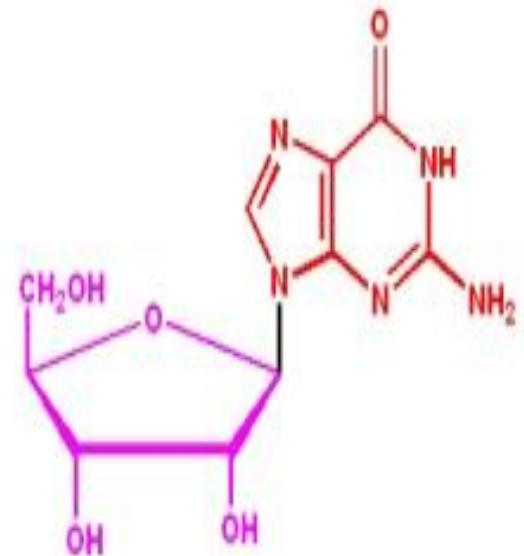
Cytidine



Uridine



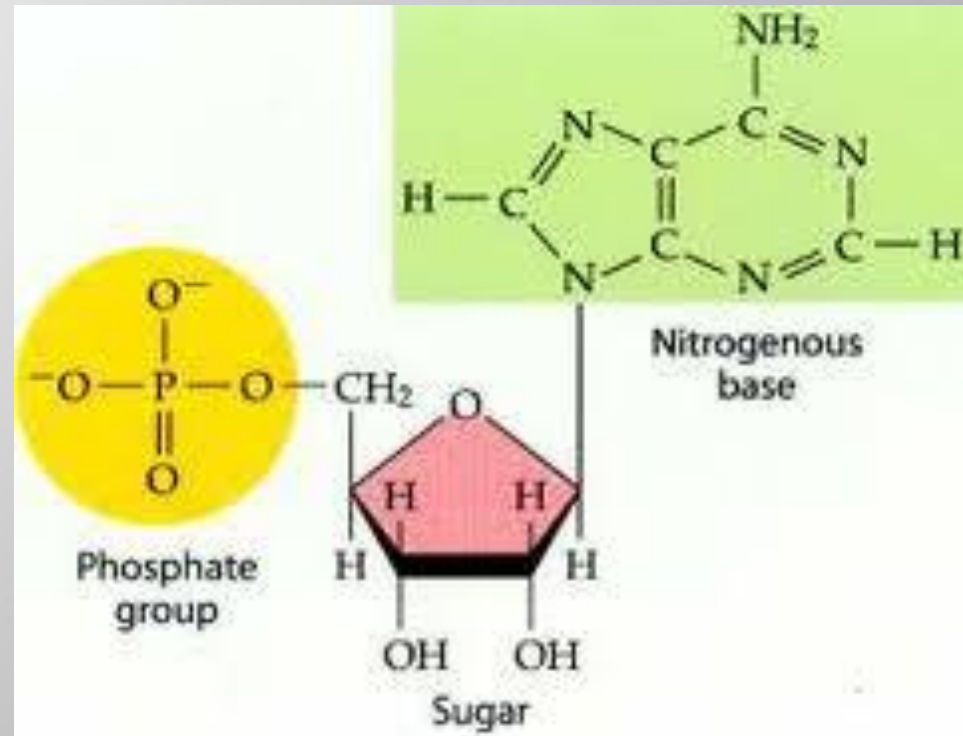
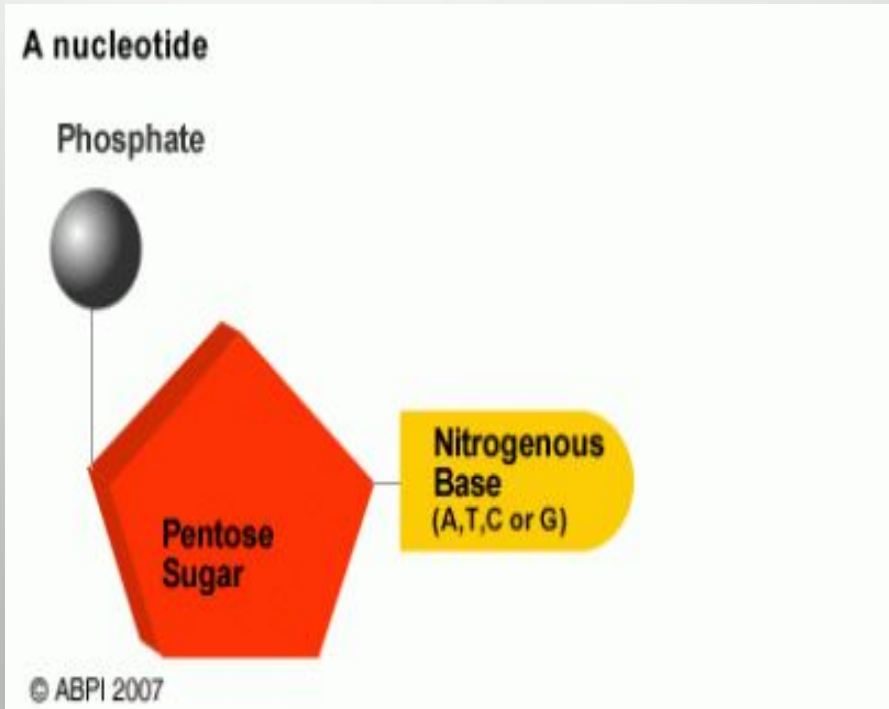
Adenosine



Guanosine

NUCLEOTIDE STRUCTURE

Nucleotide consists purine / pyrimidine base, ribose/deoxyribose and phosphates.



NOMENCLATURE

RIBONUCLEOTIDES:

- **ADENINE+ RIBOSE =ADENOSINE+ PHOSPHATE= ADENYLATE**
- **GUANINE+RIBOSE= GUANOSINE+PHOSPHATE=GUANYLATE**
- **CYTOSINE+RIBOSE=CYTIDINE+PHOSPHATE=CYTIDYLATE**
- **URACIL+RIBOSE=URIDINE+PHOSPHATE=URIDYLATE**

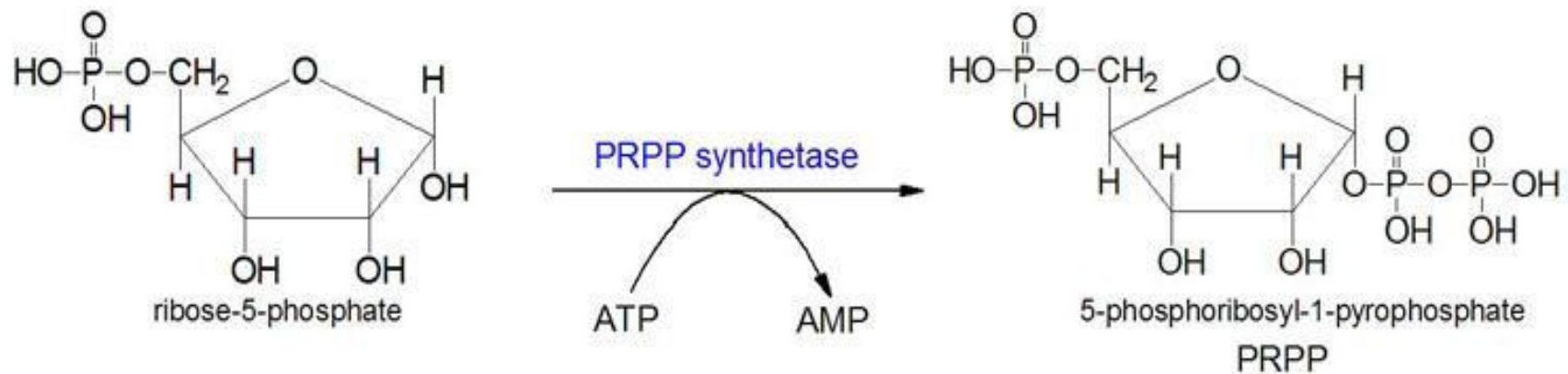
DEOXYRIBONUCLEOTIDES:

- **ADENINE+ DEOXYRIBOSE =dADENOSINE+ PHOSPHATE= DEOXYADENYLATE OR dAMP**
- **GUANINE+dRIBOSE= dGUANOSINE+PHOSPHATE= Dguanylate or dGMP**
- **CYTOSINE+RIBOSE=CYTIDINE+PHOSPHATE= CYTIDYLATE**
- **THYMINE+RIBOSE=dTHYMIDINE+PHOSPHATE=**
- **THYMIDYLATE**

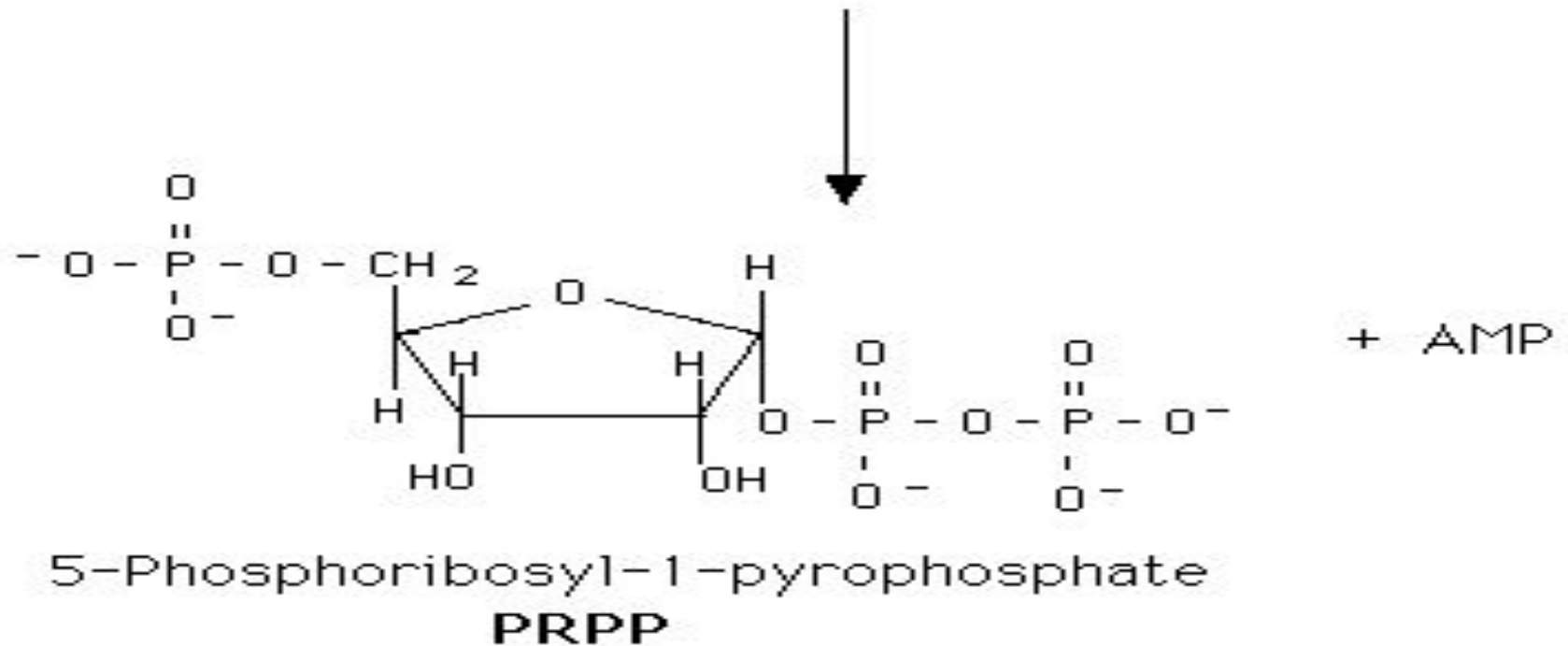
Biosynthesis of purine

It begins with PRPP (5-phosphoribosyl-1-phosphate) synthesis and PRPP synthase.

- This reaction is activated by PRPP.
- Afterwards PRPP aminotransferase synthesized 5-phosphoribosylamine.
- Those enzymes are the key regulatory enzymes for the purine synthesis.
- IMP is synthesized and could make AMP or GMP.



Ribose 5- phosphate + ATP



Ribose-5-phosphate (from pentose pathway)



5-Phosphoribosyl-1-pyrophosphate (PRPP)



5-Phosphoribosyl amine



IMP

[3,4]

AMP

ADP

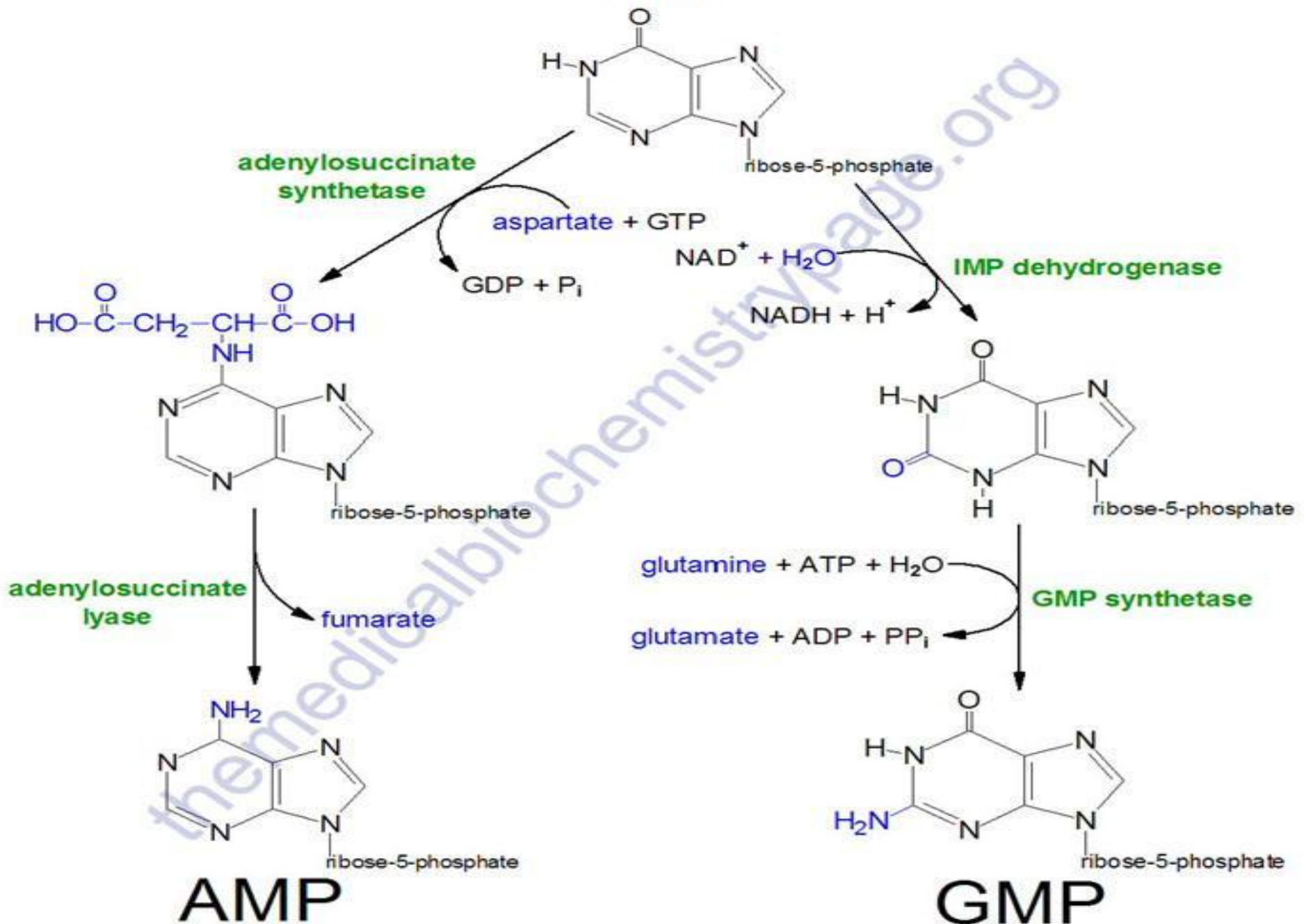
[5]

XMP

[6]

GMP

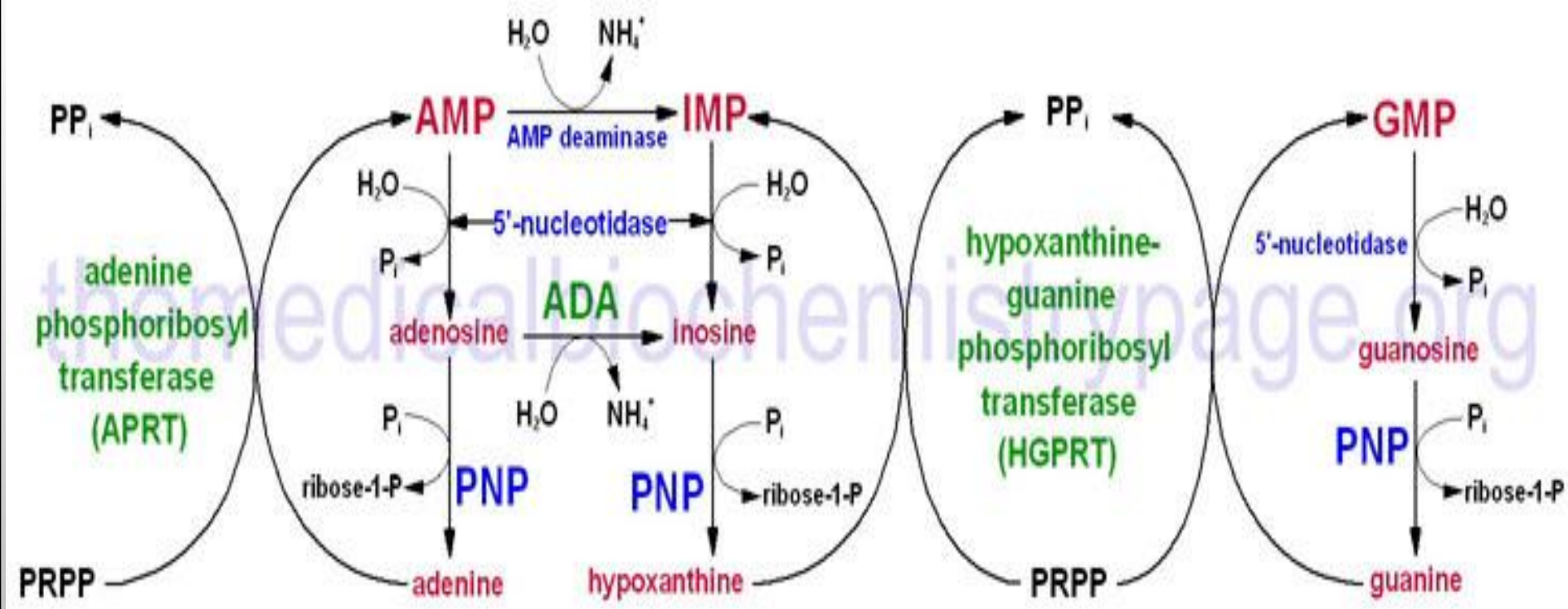
IMP



Purine salvage pathway

Liver is the major site of synthesis.

- Two enzymes are involved:
- Adenine phosphoribosyltransferase (APRT)
- Hypoxanthine-guanine phosphoribosyltransferase (HGPRT)
- It contains phosphoribosylation by PRPP to form purine mononucleotide such as adenine to AMP(with APRT) and hypoxanthine / guanine to IMP/ GMP(with HGPRT).

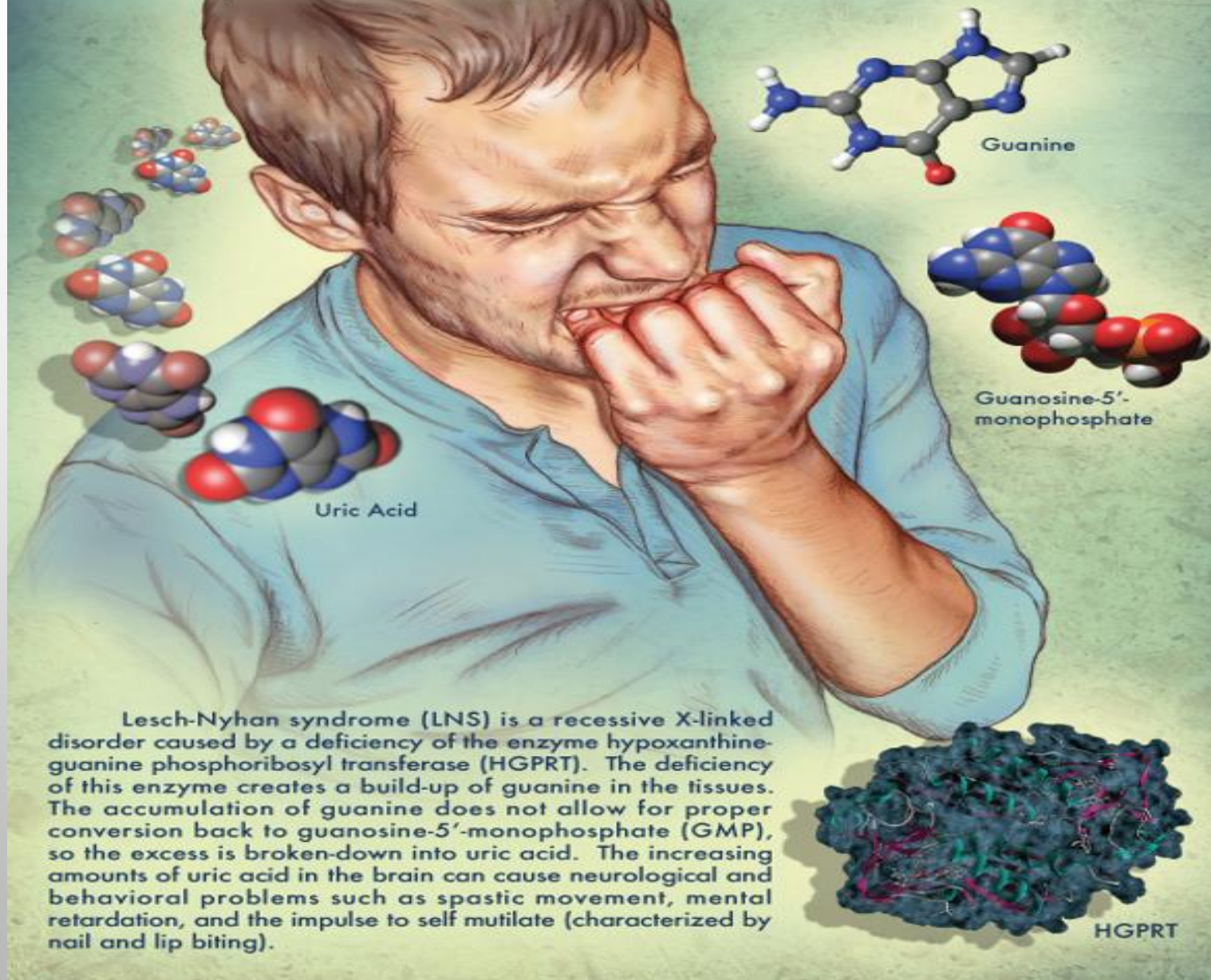


Absence of HGPRT is cause of Lesch-Nyhan syndrome

■ A rare and usually fatal genetic disorder of male children that is inherited as an X-linked recessive and is characterized by hyperuricemia, mental retardation, spasticity, compulsive biting of lips and fingers

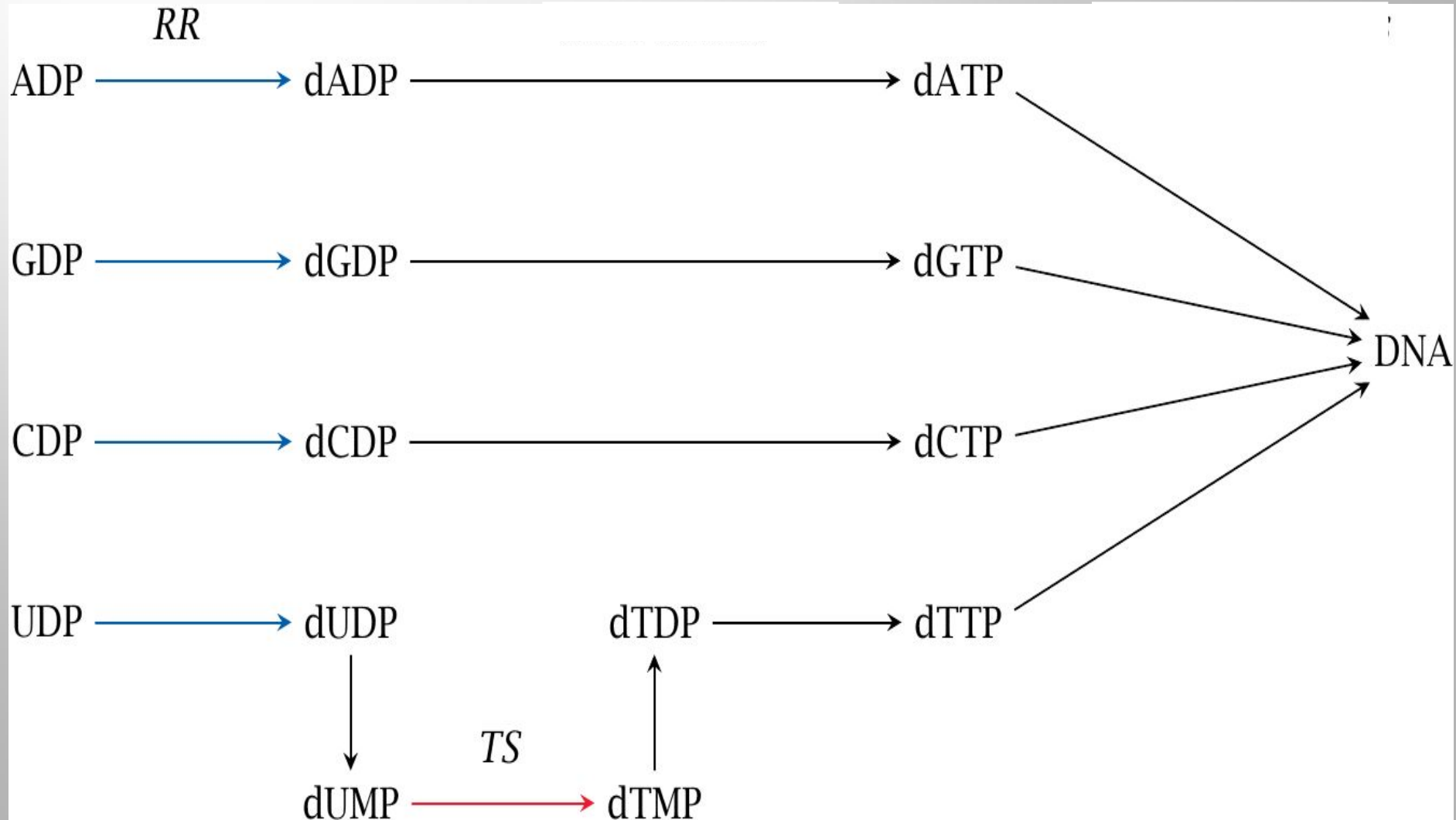


Lesch-Nyhan Syndrome



Lesch-Nyhan syndrome (LNS) is a recessive X-linked disorder caused by a deficiency of the enzyme hypoxanthine-guanine phosphoribosyl transferase (HGPRT). The deficiency of this enzyme creates a build-up of guanine in the tissues. The accumulation of guanine does not allow for proper conversion back to guanosine-5'-monophosphate (GMP), so the excess is broken-down into uric acid. The increasing amounts of uric acid in the brain can cause neurological and behavioral problems such as spastic movement, mental retardation, and the impulse to self mutilate (characterized by nail and lip biting).

SYNTHESIS OF DEOXYRIBONUCLEOTIDES



RIBONUCLEOTIDE
REDUCTASE

Ribonucleoside
diphosphate

2'-Deoxyribonucleoside
diphosphate

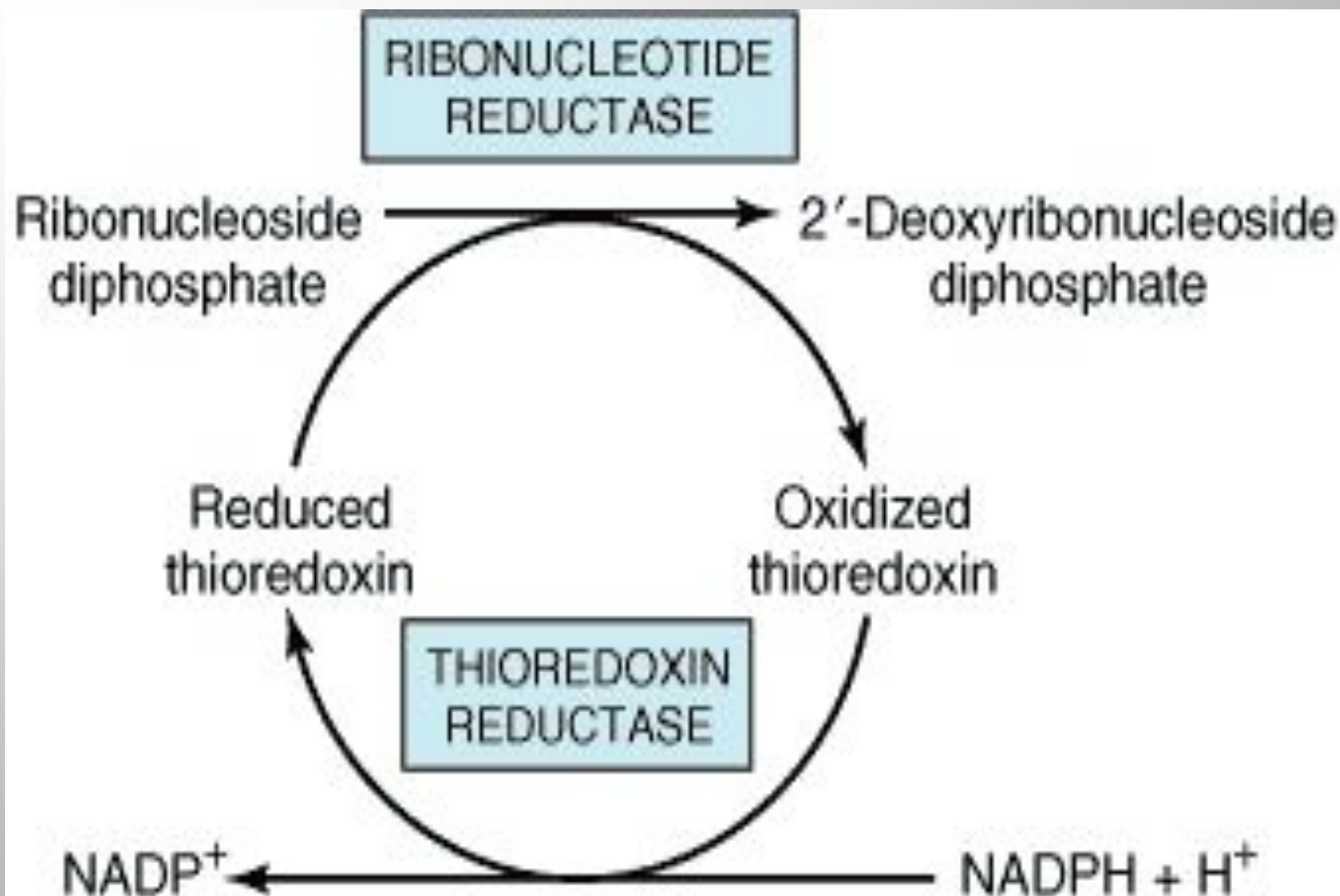
Reduced
thioredoxin

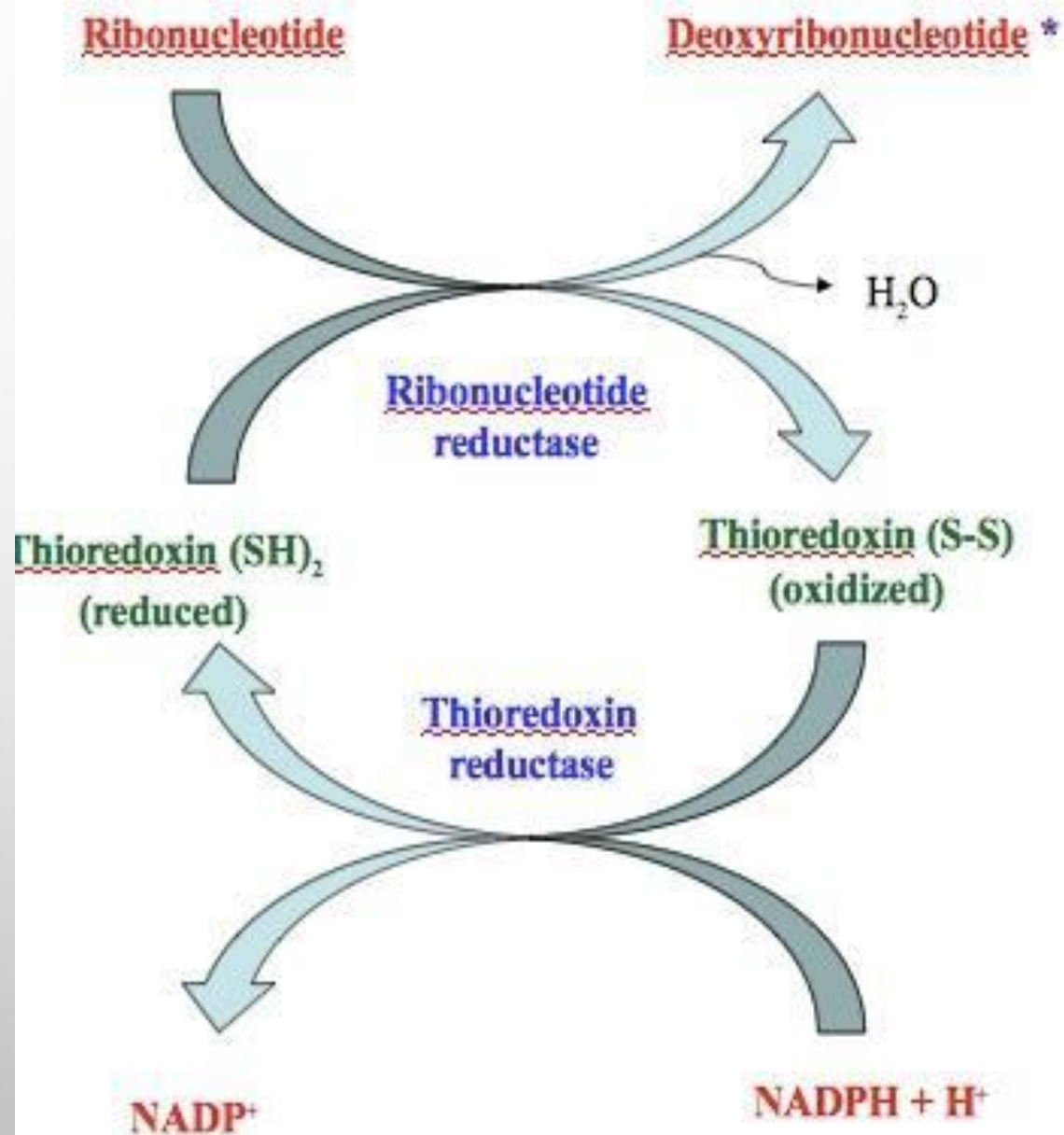
Oxidized
thioredoxin

THIOREDOXIN
REDUCTASE

NADP^+

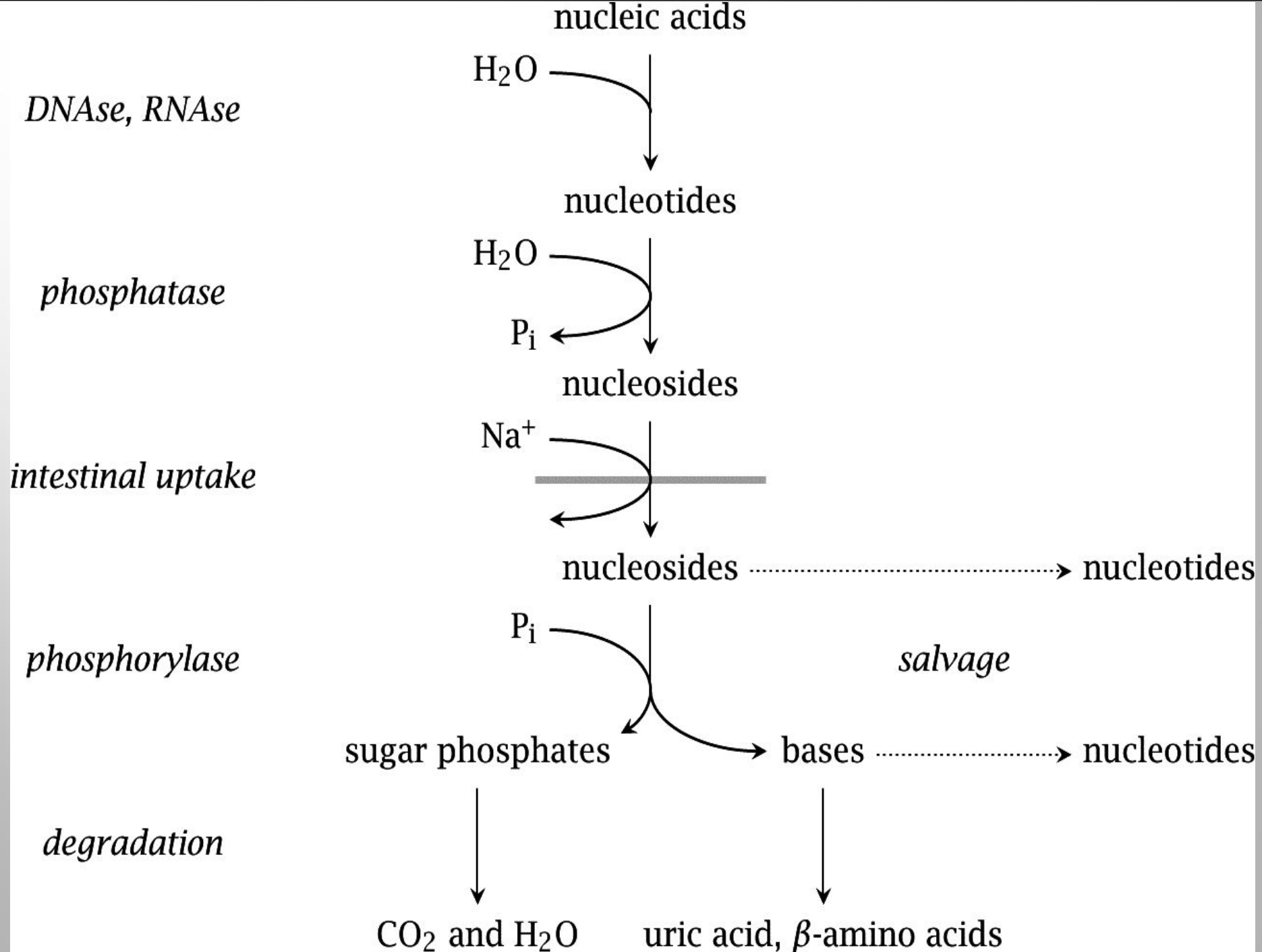
$\text{NADPH} + \text{H}^+$





Purine Catabolism

- Purines are catabolized to xanthine and uric acid in human. Uric acid then is secreted in urine.
- Salvage enzymes recycle normally 90% of these purines and 10% are converted to uric acid.

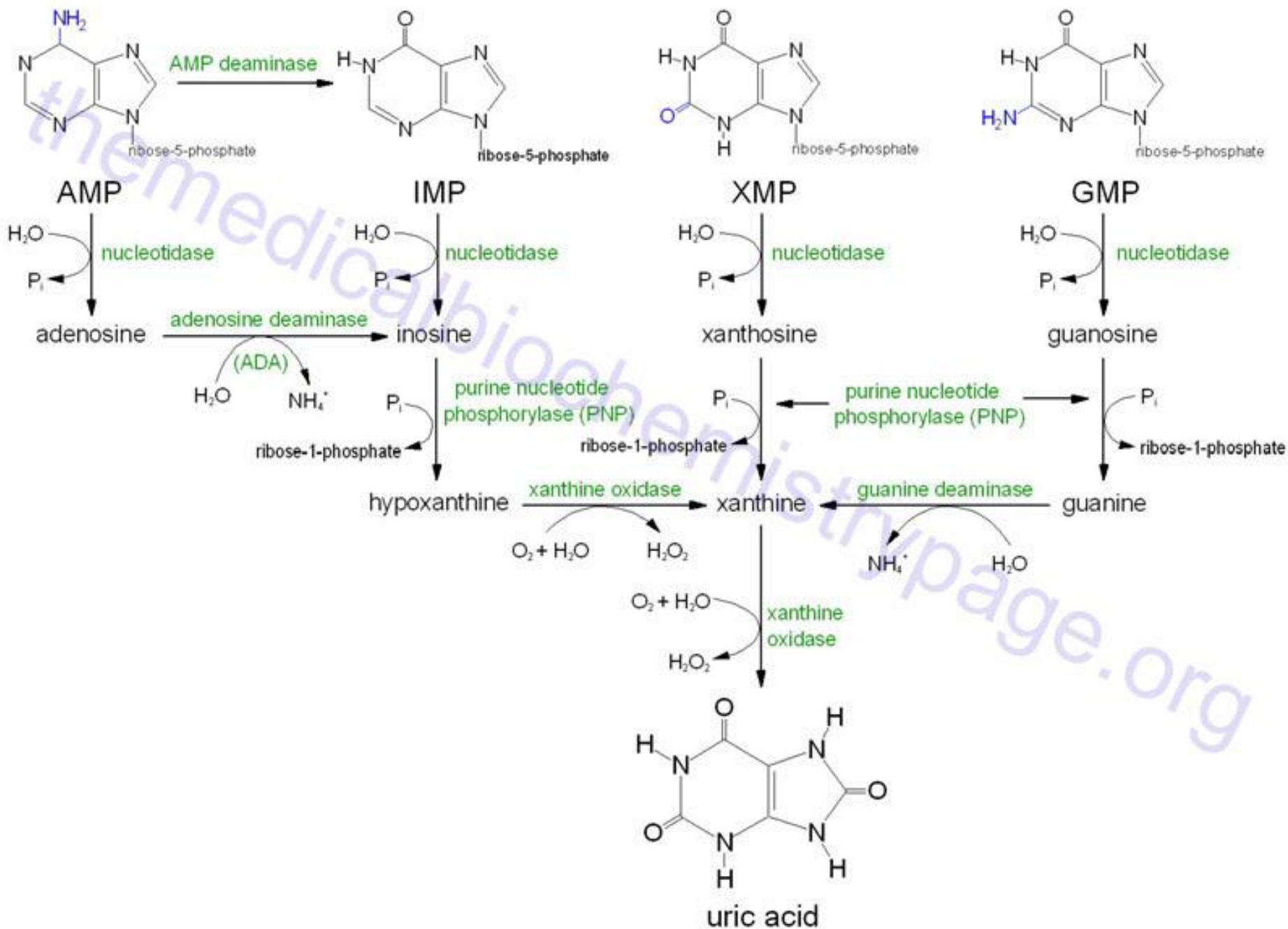


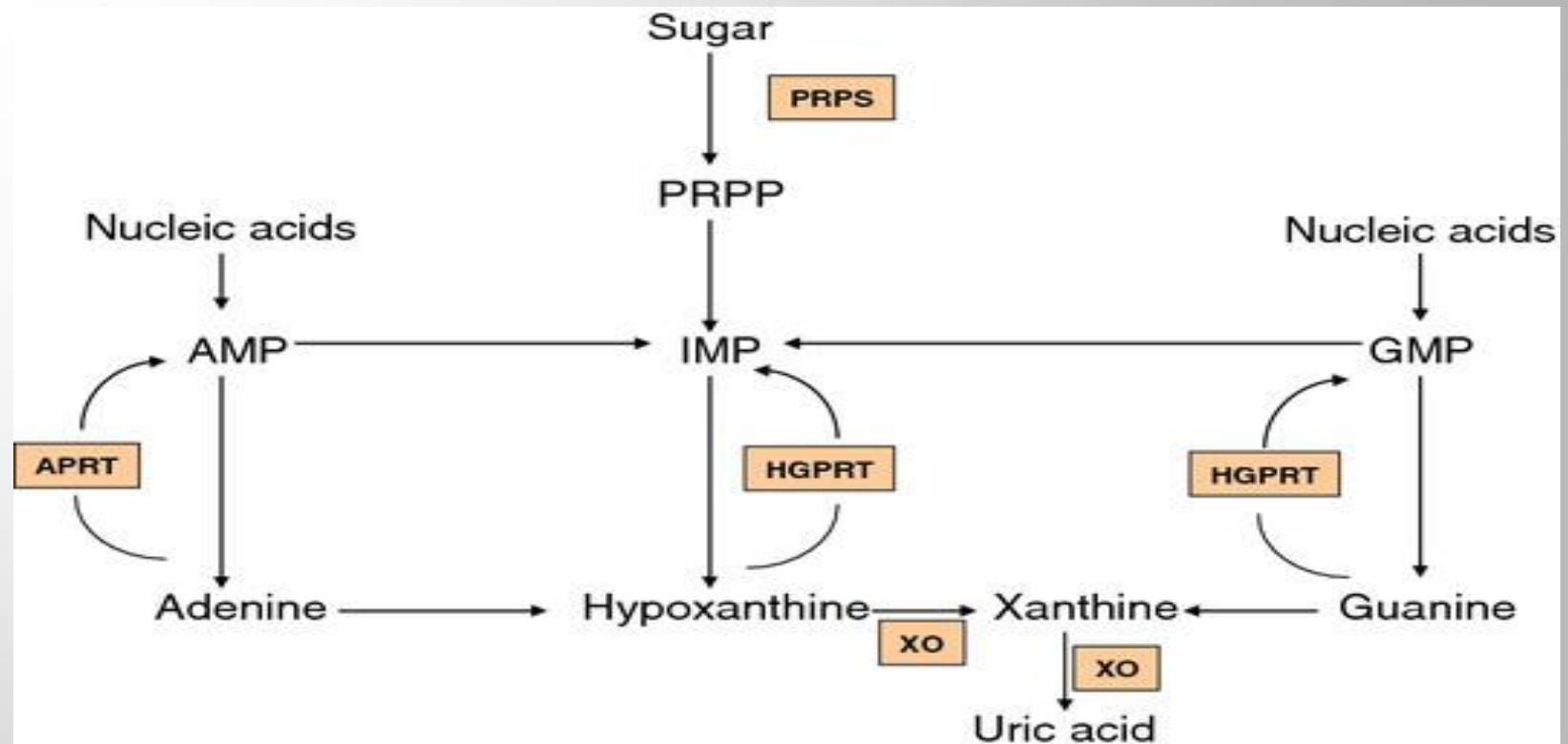
Intracellular Purine Catabolism

Nucleotides broken into nucleosides by action of 5'-nucleotidase (hydrolysis reactions)

- Purine nucleoside phosphorylase (PNP)

- Inosine □ Hypoxanthine
- Xanthosine □ Xanthine
- Guanosine □ Guanine
- Ribose-1-phosphate splits off
 - Can be isomerized to ribose-5-phosphate





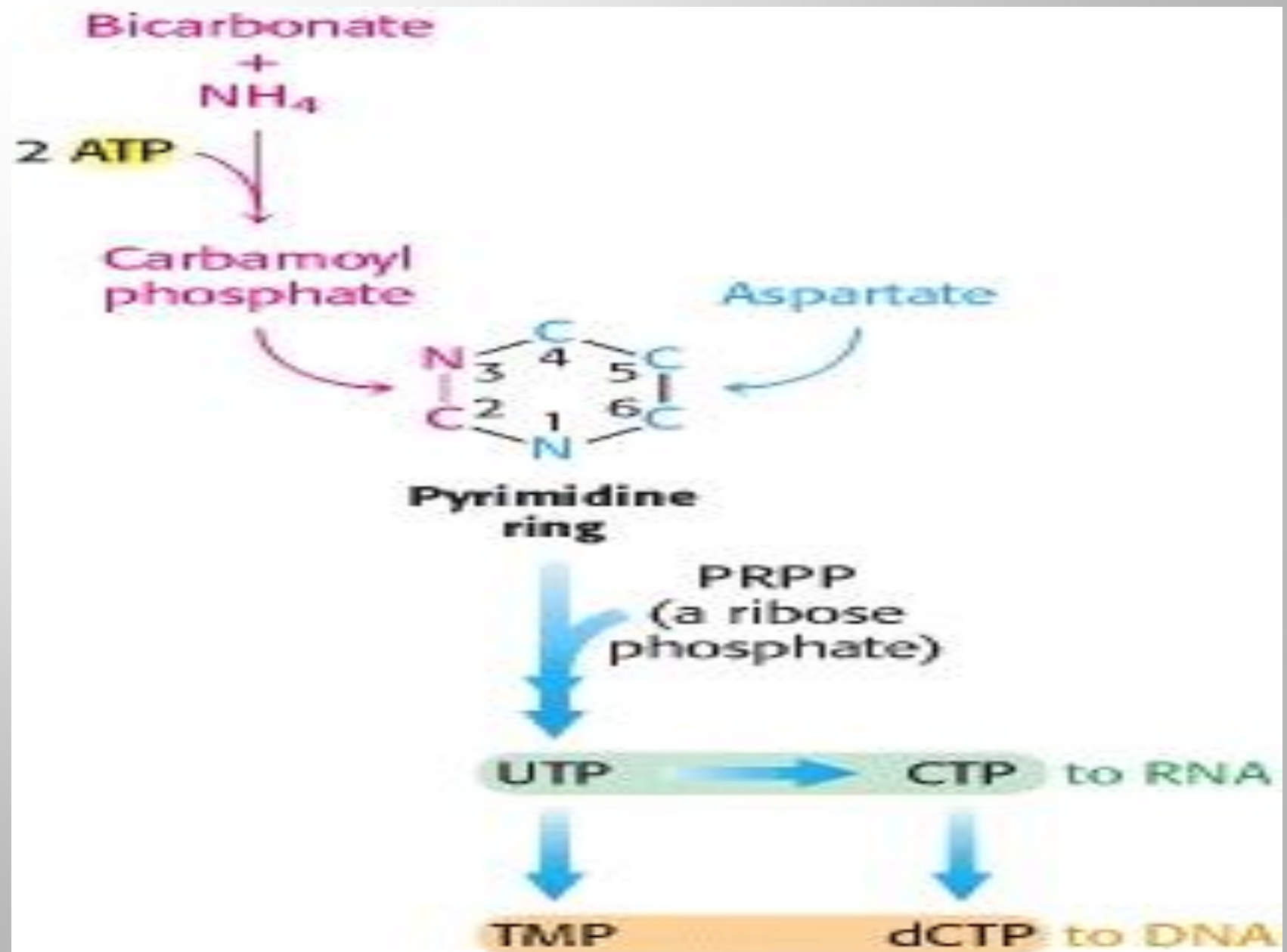
AMP	Adenosine monophosphate
APRT	Adenine phosphoribosyl transferase
GMP	Guanosine monophosphate
HGPRT	Hypoxanthine-guanine phosphoribosyl transferase
IMP	Inosine monophosphate
PRPP	5-phosphoribosyl 1-pyrophosphate
PRPS	PRPP synthetase
XO	Xanthine-oxidase

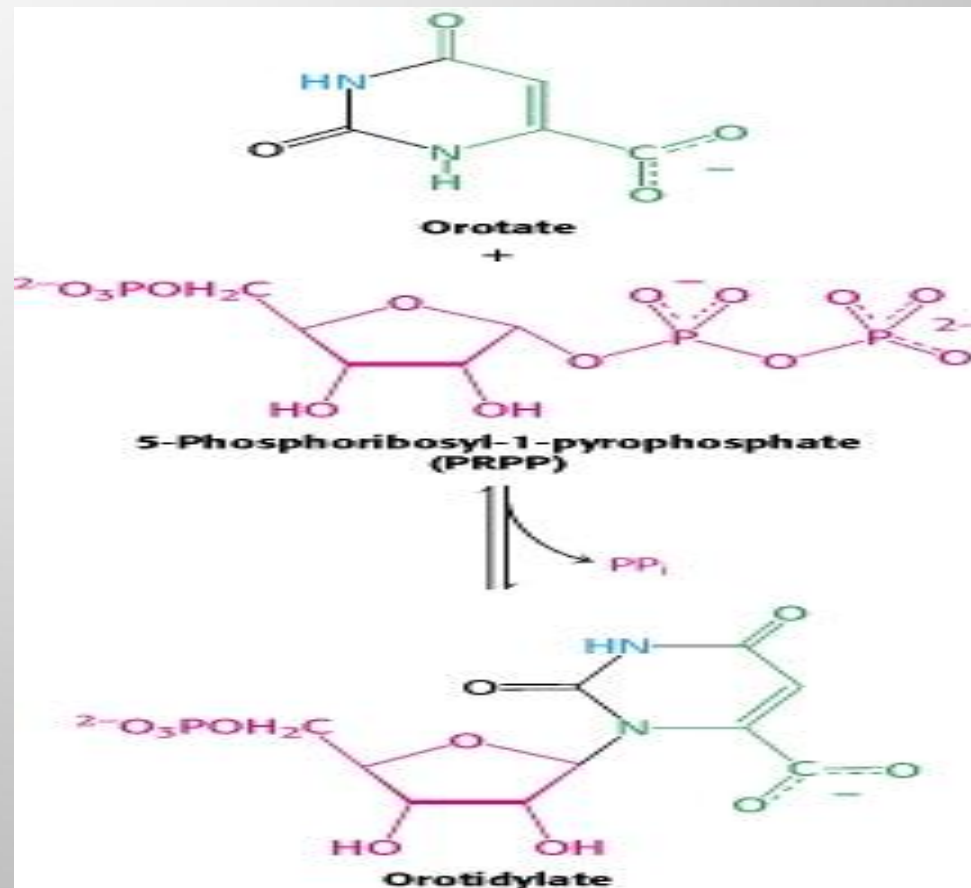
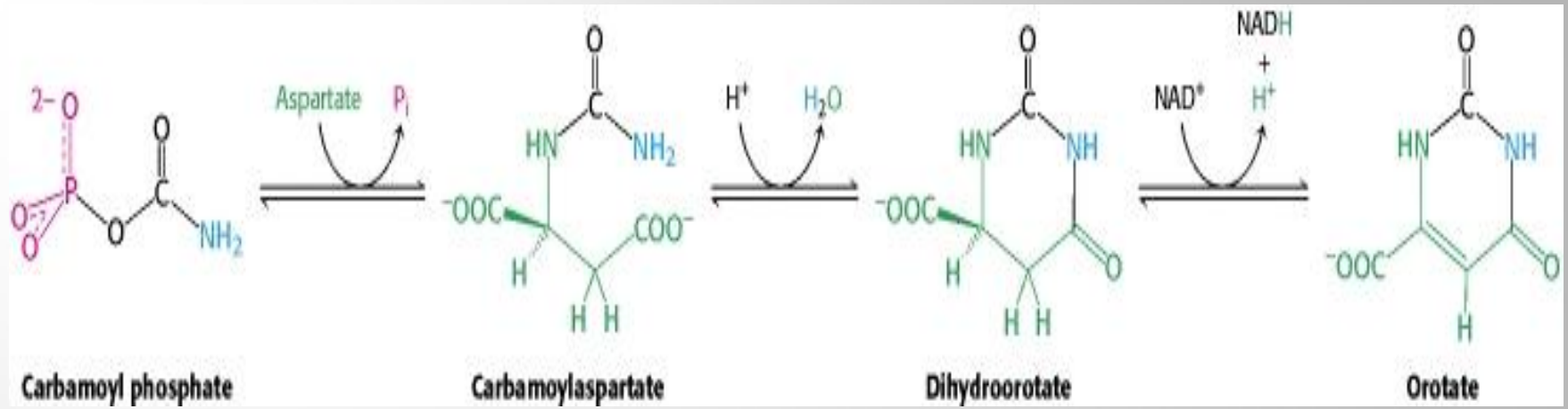
Pyrimidine synthesis

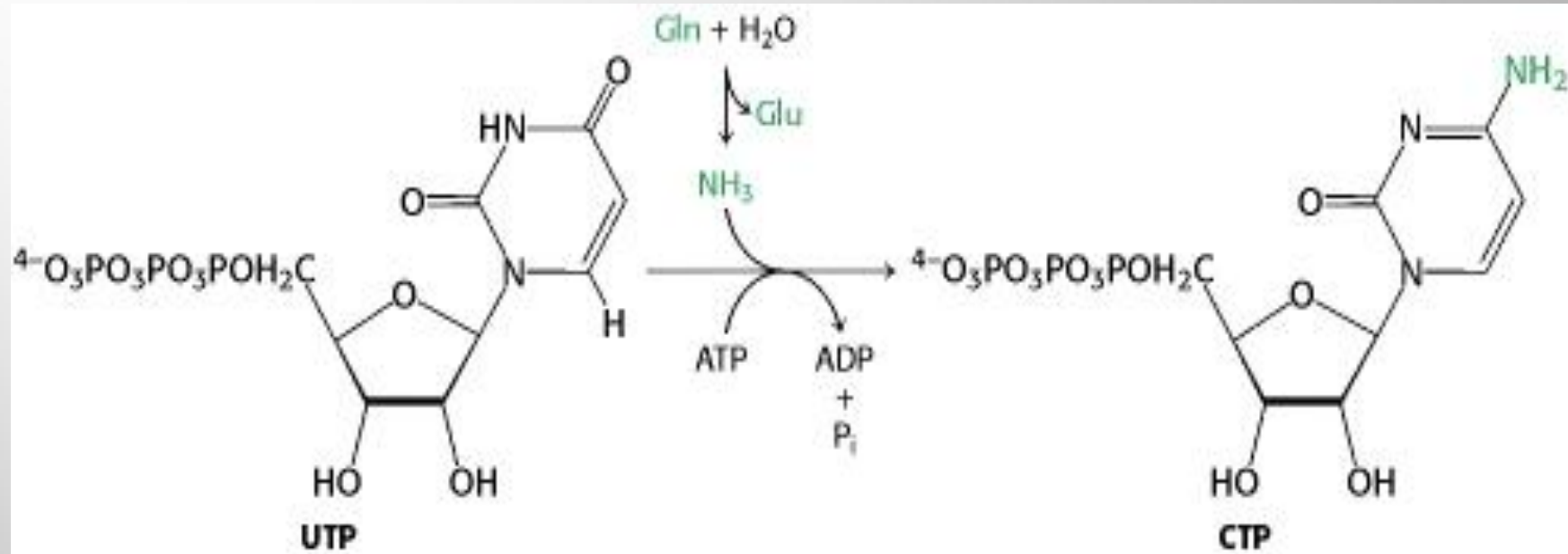
- It starts with carbamoyl phosphate with carbamoyl phosphate synthase II (in cytoplasm).
- PRPP will involve later in synthesis.
- Salvage pathway is not that important in pyrimidine synthesis.

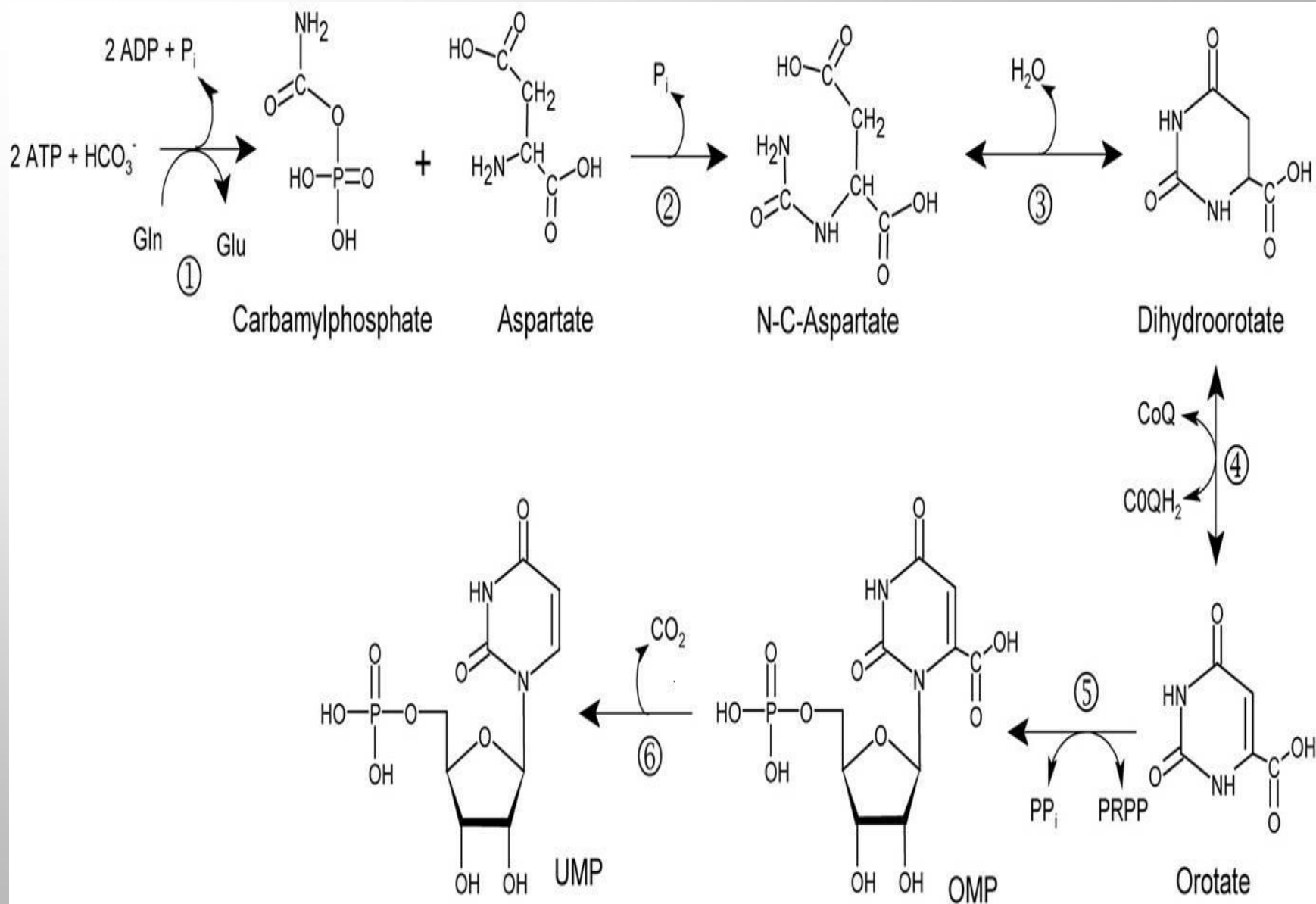
Carbamoyl phosphate synthase II is inhibited by UTP and purine nucleotides, but activated by PRPP.

- ▣ Aspartate transcarbamoylase inhibited by CTP but activated by ATP.
- ▣ The primary end product of pyrimidine synthesis is UMP.
- ▣ In the conversion of UMP to dTMP, three enzymes ribonucleotide reductase, thymidylate synthase and dihydrofolate reductase are involved.









Pyrimidine catabolism

- Pyrimidines are catabolized to β -alanine and β -aminoisobutyrate then secreted in urine.